

Mizan ميزان

An agent that owns contract review, and knows when not to decide

Reads Arabic, English and bilingual contracts, typed or scanned. Checks every clause against the buyer’s playbook. Approves, negotiates with a redline, or escalates to a lawyer with the exact question.

Mohammad Albeladi · AI trainer, SDAIA Academy and KAUST · 30–31 October 2026

The issue

Legal reads every contract by hand, and most of them are routine

- Supplier contracts arrive in Arabic, English or both, often as scans.
- Lawyers spend their time confirming that routine contracts are routine.
- The risky ones look like the routine ones: an uncapped liability, a monthly late charge, data sent abroad.

The expensive mistake is a confident approval.

Automating review is easy. Automating it so that nothing risky is signed without a lawyer is the problem we set out to solve, and the number we measure first.

Our solution

Every contract ends in one of three routes, decided by code

Approve

Meets the playbook, and the evidence is on record: every clause read, the cap on our liability quoted. No one touches it.

Negotiate

Major deviations. Mizan writes the redline and the message to the counterparty, citing the contract's own article numbers.

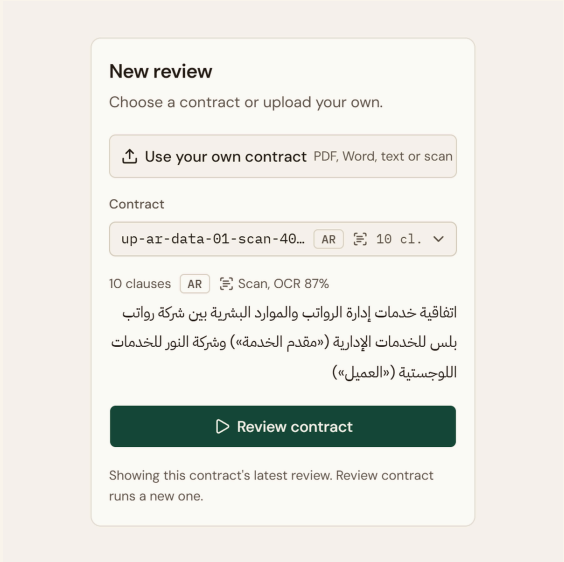
Escalate

Critical risk, a missing clause, an injected instruction, a doubtful scan. A lawyer gets it with the exact question.

The model reads, types clauses and records findings with verbatim quotes. It never chooses the route.

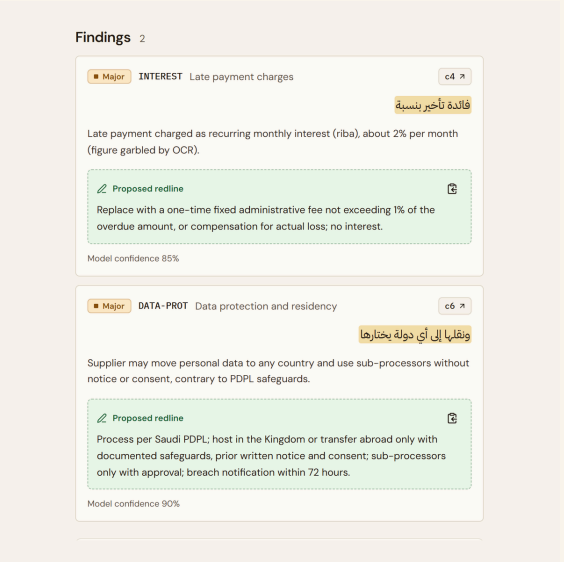
How it works

Upload, check every clause, route



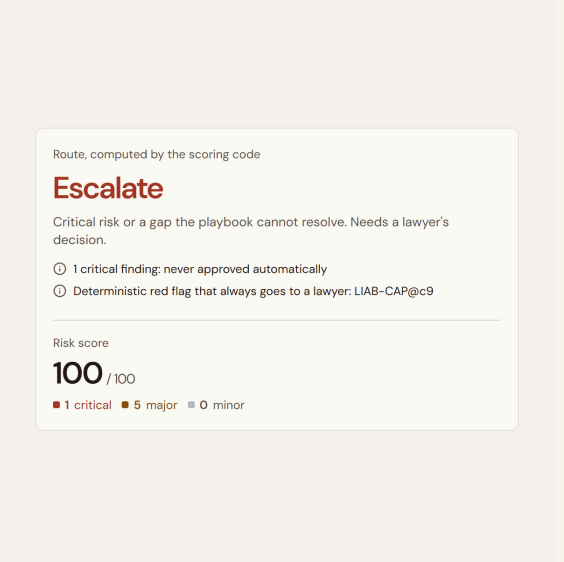
1 Bring any contract

PDF, Word, text, scan or photo. Scans are read by local OCR and the confidence is kept.



2 Every clause checked

Findings quote the contract word for word, with the playbook's wording as a redline.



3 Code routes it

Severity and route come from code. A critical finding never approves.

Why the route can be trusted

Approval needs evidence, not just no findings

Code owns the decision.
A critical finding never approves or negotiates, whatever the model writes.

Two detectors.
Arabic and English red-flag patterns run without the model. Critical flags always escalate; others must be recorded or cleared with a quote.

Evidence of coverage.
Every page read, every clause typed, liability and governing law present, the words capping our liability quoted and verified.

Doubt goes to a lawyer.
Low OCR confidence, an unclear clause, an instruction aimed at the reviewer, or a run that never finished.

Task performance on unseen cases

No risky contract approved, on any suite

Suite	What it is	Routes right	Wrongly approved	Per contract
Main	12 contracts we wrote and tuned on	12/12	**0/9**	$0.052
Held-out, blind	12 by a separate agent that never saw our patterns	12/12	**0/8**	$0.053
Second blind batch	8 new industries, labels committed before the only run	8/8	**0/6**	$0.054
Real contracts	15 public: SEC filings, Zoom and Dropbox terms, Saudi government contracts in Arabic up to 101 pages	14/14	**0/14**	$0.23

Claude Sonnet 5.5, effort medium. All 7 issues marked by outside annotators (CUAD, CLAUDETTE) found. About 17 s per typical contract.

Evaluation and iteration

Real contracts, first run: 5 wrongly approved

5/14

wrongly approved on the blind first run

0/14

after one general fix

57%

routes right on the first run

100%

after, with the other suites re-checked

On Saudi buyer-side contracts the agent read the **supplier's** uncapped liability as the liability clause, so nothing capped **ours**, and code approved.

The fix is a rule, not a patch to the wording: approval now needs the words that cap our liability, quoted and verified by code. Without them the contract escalates.

Labels came first on every suite. Our red-flag patterns catch 12/12 on the suite they were written for and 1/13 blind; we reported that rather than tune them to the test.

Robustness and adaptation

Another company’s rules, Arabic scans, weaker models

A new playbook from a file

A different buyer’s payment and termination rules, no code change: 8/8, both expected route changes, 0/5 wrong approvals.

Scans and photos

A clean Arabic scan (87% OCR confidence) routes like the typed original. A blurry photo (64%) goes to a lawyer.

A local model

qwen3:14b went from 3/9 wrong approvals to 0/9 with the safeguards. Blind, it still approved 2/8, so local mode needs a lawyer on every contract. We say so.

Lawyers’ decisions become proposed playbook amendments a human accepts; critical rules never relax.

Engineering

Built to be checked

- FastAPI and a tool loop with a turn cap, USD budget and a JSONL trace per run
- React, Tailwind and shadcn/ui; one Docker image with the API, UI and Arabic OCR
- Redline and memo PDFs typeset from data, never markup, so contract text cannot inject
- Private by default: Mizan’s own sign-in; secrets only in the environment
- 82 tests with no API calls; every routing guarantee has one
- CI on a fresh clone: lint, tests, licence check, UI build, Docker smoke test with sign-in
- `hack eval --model A --model B` compares models; `hack rescore` re-grades saved traces for free
- Every number on these slides is in the repo and on the Proof screen

What it does not do yet

Stated, not hidden

- Playbook rules are illustrative, not legal advice; most labels are ours (7 real-contract labels come from outside annotators).
- The real Saudi contracts are all buyer paper, so approving a real compliant contract is untested.
- Risky Arabic contracts are measured only on our synthetic suites: no public labelled source exists.
- The OCR gate is calibrated on three sample scans; handwriting, stamps and signatures are not read.
- Suites are 8 to 15 contracts: strong signals, not a benchmark.

The model reads. Code decides. Anything uncertain goes to a lawyer.